name: Matrix Test

on: push

jobs:
 test:
  runs-on: ubuntu-latest

strategy:
   matrix:
    python-version: [3.8, 3.9, 3.10]

steps:
   - uses: actions/checkout@v4

- uses: actions/setup-python@v5
    with:
     python-version: ${{ matrix.python-version }}

- run: python --version explain

This workflow uses a **matrix strategy** to test your code on **multiple Python versions** in parallel.

What This Workflow Does

1. Trigger

```
on: push
```

- Runs on every push to any branch

2. Matrix Strategy

```
strategy:
  matrix:
    python-version: [3.8, 3.9, 3.10]
```

- Creates 3 **separate jobs** running in parallel:
  - One with Python 3.8
  - One with Python 3.9

- One with Python 3.10

3. Steps Breakdown

Step	What It Does
`actions/checkout@v4`	Clones your repository so the code is available [1] [2]
`actions/setup-python@v5`	Installs the Python version from the matrix (`${{ matrix.python-version }}`)
`python --version`	Prints the Python version to verify it's correct

What You'll See in GitHub Actions UI

When you run this workflow, you'll see **3 separate job runs** under the `test` job:

```
test (3.8)  ✅
test (3.9)  ✅
test (3.10) ✅
```

Each job's output will show:

```
Python 3.8.x
Python 3.9.x
Python 3.10.x
```

Key Concepts for Interview

Concept	Explanation
Matrix Build	Runs the same job with different variable combinations [3]
`${{ matrix.python-version }}`	Dynamic reference to current matrix value
Parallel Execution	All 3 versions run simultaneously (faster than sequential)
`runs-on: ubuntu-latest`	All jobs use Ubuntu runner (you could also matrix over OS)

Common Matrix Extensions

Add more dimensions:

```
strategy:
  matrix:
    python-version: [3.8, 3.9, 3.10]
    os: [ubuntu-latest, macos-latest]
```

→ Creates 6 jobs (3 versions × 2 OS)

Exclude specific combinations:

```
strategy:
  matrix:
    python-version: [3.8, 3.9, 3.10]
    exclude:
      - python-version: 3.8
        os: macos-latest
```

Include additional values:

```
strategy:
  matrix:
    python-version: [3.8, 3.9]
    include:
      - python-version: 3.11
        experimental: true
```

Why This Matters in Real DevOps Work

- Ensures your code works across **supported Python versions**
- Catches version-specific bugs early
- Required for publishing packages to PyPI (must support multiple versions)
- Common pattern for CI in open-source Python projects

⁂

1. https://notes.kodekloud.com/docs/GitHub-Actions-Certification/GitHub-Actions-Core-Concepts/Configure-Checkout-Action
2. https://dev.to/spacelift/using-checkout-action-in-github-actions-workflow-2bna
3. https://docs.github.com/actions/using-workflows/events-that-trigger-workflows